

Knuth's Optimization

Knuth's optimization, also known as the Knuth-Yao Speedup, is a special case of dynamic programming on ranges, that can optimize the time complexity of solutions by a linear factor, from $O(n^3)$ for standard range DP to $O(n^2)$.

Conditions

The Speedup is applied for transitions of the form

$$dp(i, j) = \min_{i \leq k < j} [dp(i, k) + dp(k+1, j) + C(i, j)].$$

Similar to [divide and conquer DP](#), let $opt(i, j)$ be the value of k that minimizes the expression in the transition (opt is referred to as the "optimal splitting point" further in this article). The optimization requires that the following holds:

$$opt(i, j-1) \leq opt(i, j) \leq opt(i+1, j).$$

We can show that it is true when the cost function C satisfies the following conditions for $a \leq b \leq c \leq d$:

1. $C(b, c) \leq C(a, d)$;
2. $C(a, c) + C(b, d) \leq C(a, d) + C(b, c)$ (the quadrangle inequality [QI]).

This result is proved further below.

Algorithm

Let's process the dp states in such a way that we calculate $dp(i, j-1)$ and $dp(i+1, j)$ before $dp(i, j)$, and in doing so we also calculate $opt(i, j-1)$ and $opt(i+1, j)$. Then for calculating $opt(i, j)$, instead of testing values of k from i to $j-1$, we only need to test from $opt(i, j-1)$ to $opt(i+1, j)$. To process (i, j) pairs in this order it is sufficient to use nested for loops in which i goes from the maximum value to the minimum one and j goes from $i+1$ to the maximum value.